

Fig. 3: Actual-size PCB of the metal detector

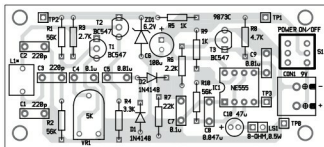


Fig. 4: Component layout of the PCB

rectified and filtered. Resulting DC voltage is given to switching transistor T3.

Under normal conditions, when there is no object with iron contents near sensing coil L1*, the oscillator generates RF signals, transistor T3 conducts and keeps reset pin 4 of IC1 to ground level. There will be no output at pin 3 of IC1; accordingly, no sound comes in from the speaker.

But when an object with iron contents is brought near (within 2.5cm) sensing coil L1*, the oscillator stops oscillating. Transistor T3 stops conducting and, as a result, reset pin 4 of IC1 is held high. IC1 goes into astable mode. Output at pin 3 is in audio range, which can be heard through the speaker.

Air-core coil L1* has 20 turns of 26SWG enamelled copper wire having a diameter of 6cm. There is no former. Refer to Fig. 2 before winding L1*.

Construction and testing

An actual-size, single-side PCB for the metal detector is shown in Fig. 3 and its component layout in Fig. 4. Enclose the PCB in a suitable small box in such a way that the object under scanning is brought within 2.5cm of the sensing coil. Ensure proper wiring to avoid any mistake.

Verify that voltages at the test points are as per Test Points table before using the circuit. The circuit works on 9V battery or 9V regulated power supply. ●



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